Design & Aerodynamic Justification Report: FWUAV-MK1

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Target Mission Profile: Achieve successful cruise flight

Executive Summary

- **Mission Goal:** Achieve cruise flight at low speed
- **Key Design Outcomes:** Lightweight, Stable, durable.
- **Performance Summary:** Cruising speed = 15 m/s, maximum takeoff weight (MTOW) = 1kg, high lift-to-drag ratio $(L/D)_{max}$, and stall speed as low as possible.

1. Mission Requirements & Operational Constraints

Establish the baseline specifications that dictate the aerodynamic configuration.

Parameter	Design Value	Rationale
Maximum Takeoff Weight (MTOW)	1 kg	Battery + Payload + Airframe estimate
Cruise Speed (V_{cruise})	15 m/s	Easy to control UAV for constrained flight area
Stall Speed (V_{stall})	5.397 m/s	Safe hand-launch / short runway takeoff requirement
Max Payload Weight	No payload	-

2. Aerodynamic Layout & Configuration Selection

2.1 Overall Configuration

- **Selected Architecture:** Fixed rectangular flat wing
- **Justification:** Conventional rectangular flat wing is chosen over dihedral or delta wing configuration as to easily fabricate the UAV, avoiding complexity

2.2 Propulsion Integration

- **Selection:** Tractor configuration (propeller fixed at front of UAV)
- **Justification:** In a case of nose dive, motor takes the hit instead of fuselage, easy to repair and maintain.

3. Wing & Airfoil Aerodynamic Design

This is the core technical section of your aerodynamic justification.

3.1 Airfoil Selection

- **Candidate Airfoil Evaluated:** The SD7037 delivers high lift-to-drag and stable boundary-layer attachment, mitigating laminar separation bubbles and early stall.
- **Operating Reynolds Number (Re):**
  $Re = (\rho \cdot V \cdot c) / \mu$
  $\rho = 1.225$ kg/m^3
  v = 15 m/s (required velocity)
  c = 35 cm
  $\mu = 1.8 * 10^{(-5)}$ Pa · s
- **Justification Criteria:**
  - 120 cm of maximum wingspan
  - 35 cm chord length as to maximize wing area without compromising Aspect Ratio
- **Selection:** SD 7037

3.2 Wing Geometry & Planform Design

- **Aspect Ratio (AR):**
  AR = b^2/s
  = b/c (for rectangular wing)
  AR = 3.42
- **Wing Loading:**
  Wing Loading = W/S = 1kg/0.42m^2 = 2.38 kg/m^2
- **Wing Taper Ratio (λ):** 1 for rectangular wing
- **Sweep Angle (Λ):** Typically 0° for low-speed UAVs
- **Dihedral / Anhedral Angle:** 0° as dihedral angle is not required for the mission

4. Stability, Control, and Empennage Design

4.1 Longitudinal & Lateral-Directional Stability

- **Control Surfaces (Ailerons, Elevator, Rudder):**
  - Aileron = Approximately 15% of the wing area.
  - Elevator and Rudder = Approximately 25% of the HT and VT respectively.

5. Drag Breakdown

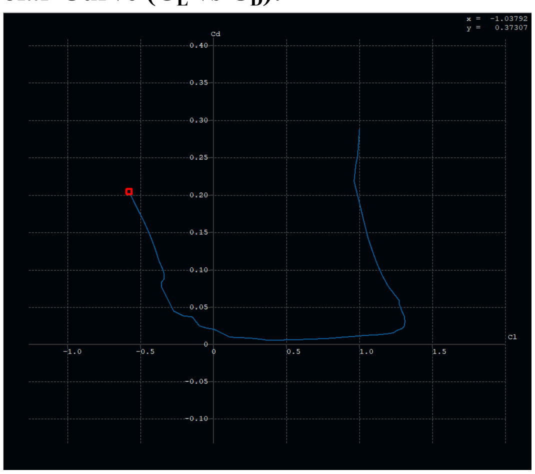
5.1 Parasite & Induced Drag Estimation

Break down drag components at cruise angle of attack:

- **Zero-Lift Drag (C_{D0}):** 0.02116
- **Induced Drag (C_{Di}):**
  C_{Di}= kCL^2 (k= 1/π · e · AR)
  Where e = oswalds efficiency factor
- **Total Drag Coefficient (C_D):**
  $C_D = C_{D0} + (C_L^2 / (\pi \cdot e \cdot AR))$
- **Oswalds efficiency factor (e):**
  $e = 1.78 \times (1 - 0.045 \times AR^{0.68}) - 0.64 = 0.955 = 95.5\%$ **-** using empirical formula

6. Performance Parameters

6.1 Performance Polars

- **Polar Curve (C_L vs C_D):**
  

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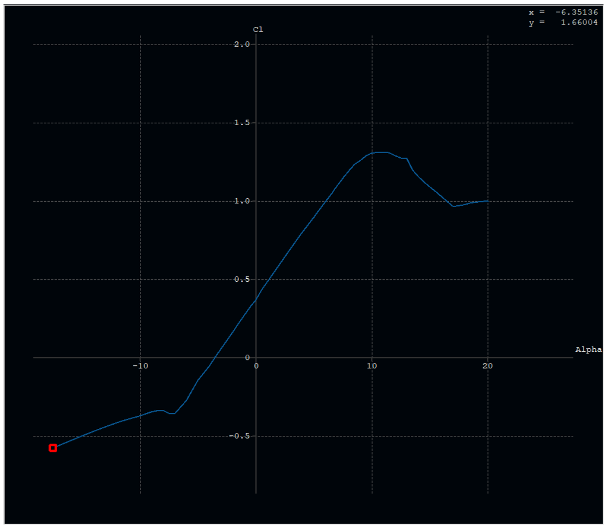
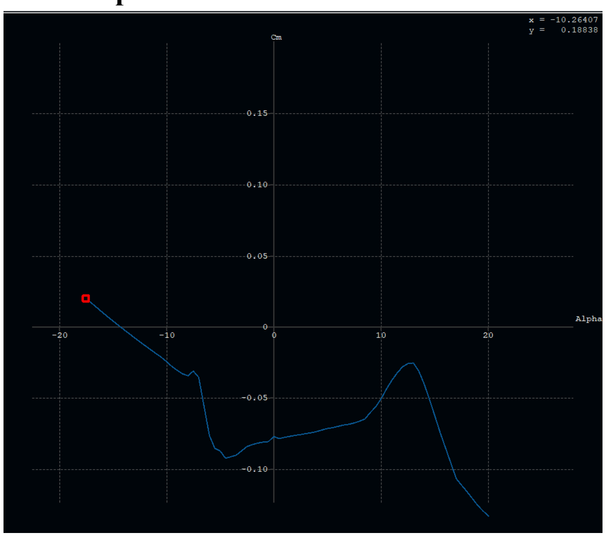
- **Glide Ratio / Efficiency:**
  $$(L/D)max \quad = \quad \frac{1}{\sqrt{4kC_{D0}}} = 11.02247$$
- **CL vs Alpha:**



- **Cm vs Alpha:**



6.2 Performance Criterias:

- **Minimum Thrust for Cruise :**
  Treq = W √ 4kCD0 = 0.89N
- **Minimum Cruise Velocity:**
  Velocity corresponding to Trmin = 9.04 m/s
- **Stall Velocity:**
  Velocity corresponding to stall angle = 5.397 m/s
- **Minimum Power Required for Cruise:**
  Preq = 7.07147 W
- **Power required considering loss of power at propulsion**
  Pprop = nshaft * Pshaft
  Hence
  Pshaft = 10W (assuming 75% efficiency by propeller used)

7. Computational & Experimental Validation

7.1 XFLR5 Analysis:

- **Software Used:** XFLR5
- **Analysis Conditions:** Angle of attack (α) sweep from -20° to 20°.

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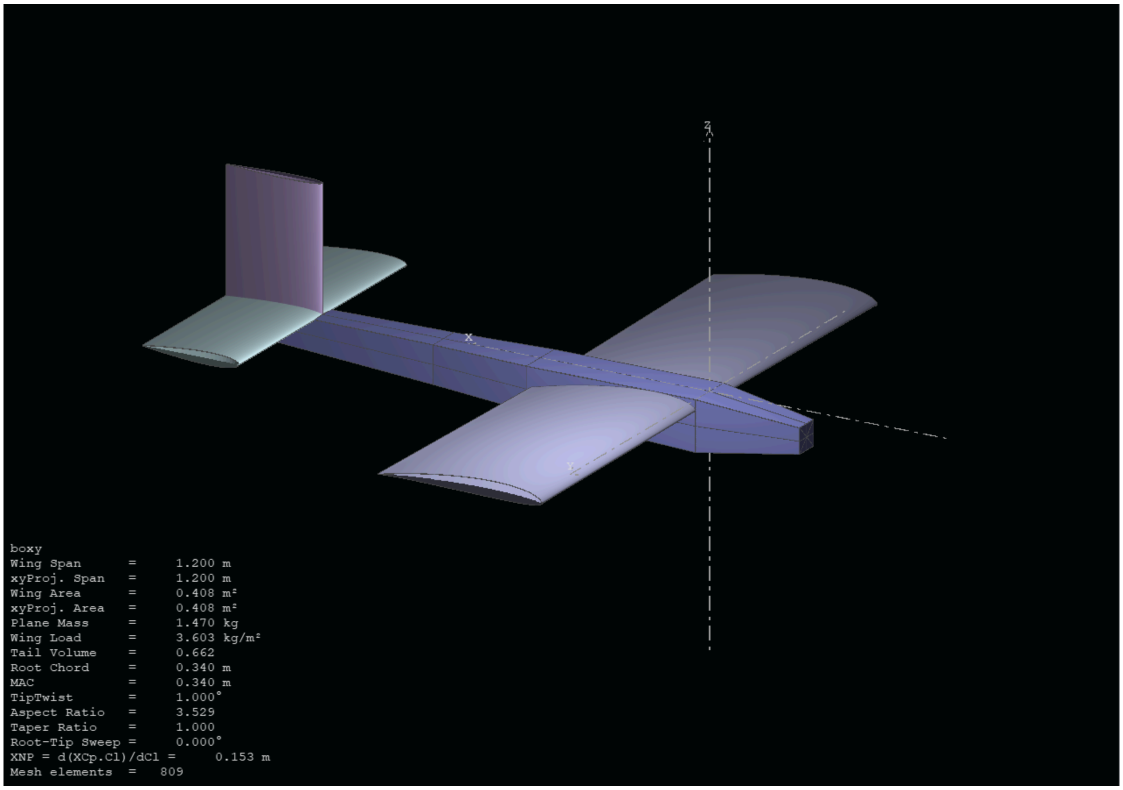
8. CAD/Design of Prototype



9. Fabrication

9. Conclusion & Recommendations

- **Succesfully achieved cruise flight**
- **Future Improvements:** Potential Usage of UAV telemetry unit, FPV cameras, could use better options than depron + cardboard + wood

*Video of Test flight can be seen on the github repository